RECOMMENDATION

UnitedHealth Group (UNH): Long

Base-case DCF value $526.64 per share against an as-of price of $367.08: +43.5%.

$354.47	$526.64	$764.68
bear (-3.4%)	base (+43.5%)	bull (+108.3%)

As of 2026-09-30. The value is an intrinsic-value estimate, not a dated price target. The bear case sits *below* the price, and the result depends on a low beta (slide 11). Educational project, not investment advice.

WHY

Thesis in three bullets

1. **The price assumes little growth.** The reverse DCF says the price implies `6.0%` stage-1 FCF growth. UNH's median rolling 5-year FCF growth is `13.7%`.
2. **Revenue kept growing.** Revenue grew from `$287.6B` (FY2021) to `$447.6B` (FY2025). The problem is margin, not demand.
3. **The thesis is a margin recovery, and it is not yet in the numbers.** Operating margin fell from `8.7%` to `4.2%`. The long case needs FCF to stop shrinking.

FUNDAMENTALS

Business snapshot

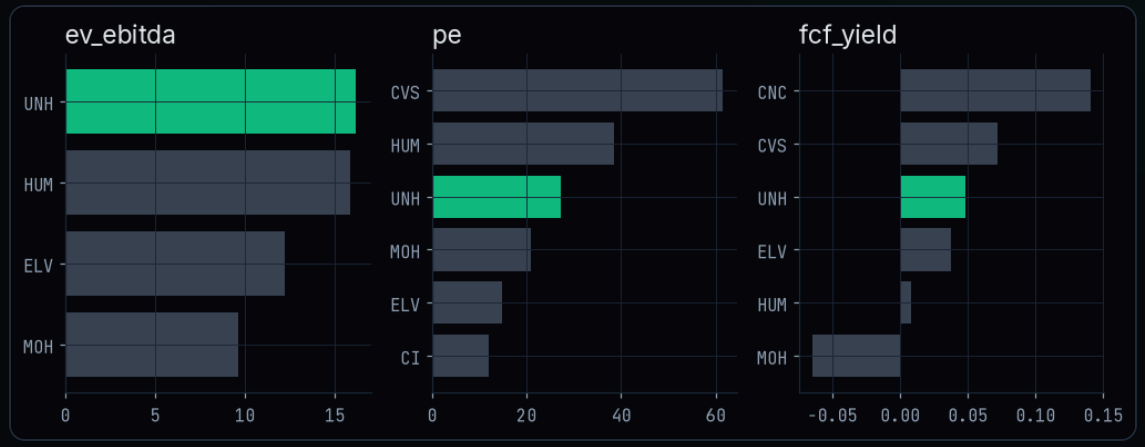
Fiscal year	Revenue	Operating income	Op. margin	FCF	FCF margin
FY2021	$287.6B	$24.0B	8.3%	$19.9B	6.9%
FY2022	$324.2B	$28.4B	8.8%	$23.4B	7.2%
FY2023	$371.6B	$32.4B	8.7%	$25.7B	6.9%
FY2024	$400.3B	$32.3B	8.1%	$20.7B	5.2%
FY2025	$447.6B	$19.0B	4.2%	$16.1B	3.6%

SEC EDGAR 10-K figures as first reported. FCF = operating cash flow − capex. Revenue keeps rising while operating income and FCF shrink. That gap is the key driver to watch.

RELATIVE VALUE

Comps: a premium on earnings, a discount on cash

	EV/EBITDA	P/E	FCF yield	Gaps
UNH	16.2x	27.3x	4.9%	
ELV	12.2x	14.9x	3.8%	
CI	n/a	12.1x	n/a	capex not reported in XBRL; total debt not reported in XBRL
HUM	15.9x	38.6x	0.8%	
CNC	n/a	n/a	14.1%	net loss: P/E not meaningful; negative EBITDA: EV/EBITDA not meaningful
MOH	9.6x	20.9x	-6.4%	
CVS	n/a	61.5x	7.2%	total debt not reported in XBRL
Peer median	12.2x	20.9x	3.8%	

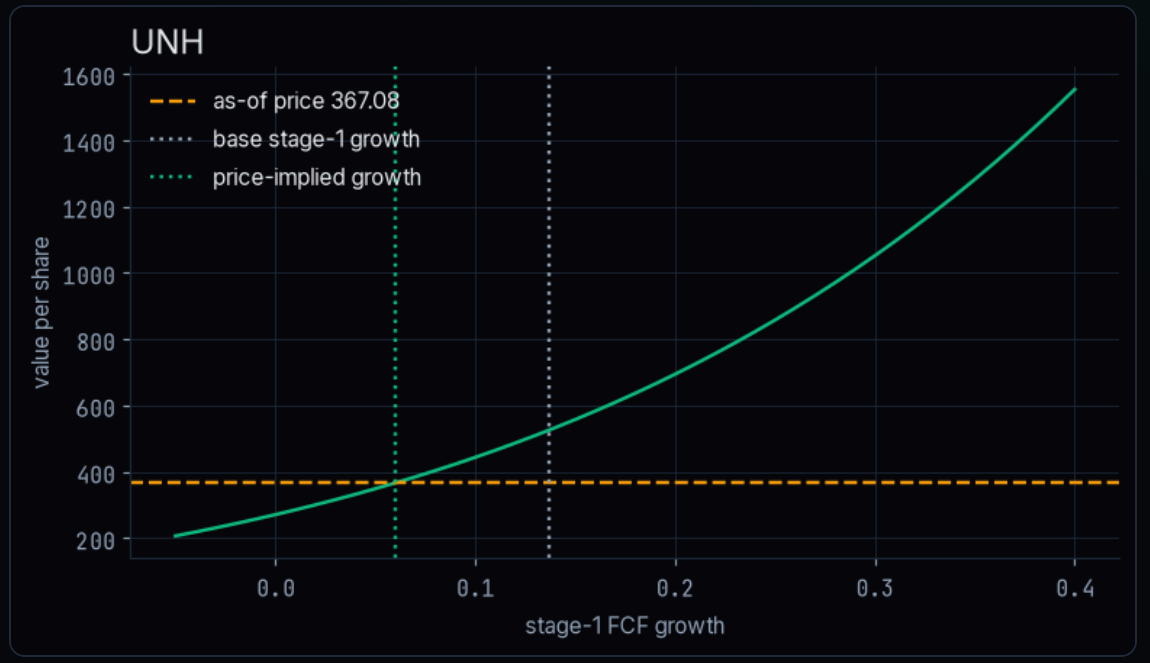


UNH trades above the peer median on P/E and EV/EBITDA but offers a higher FCF yield. Comps alone don't make the case. Gaps are left blank, never estimated.

VALUATION

DCF and reverse DCF: what the price assumes

risk free	5.3%	FRED DGS10 on 2026-09-30
beta	0.41	5y daily stock vs SPY returns to as-of (OLS slope)
equity risk premium	8.3%	Ken French Mkt-RF, arithmetic mean 1926-07 to 202608, x12
cost of equity	8.7%	CAPM: rf + beta x ERP
pre tax cost of debt	5.5%	FY2025 interest expense / average total debt (SEC 10-K)
tax rate	12.9%	FY2025 income tax / pretax income (SEC 10-K)
equity weight	82.0%	market cap (as-of price x Yahoo shares) / (market cap + book debt)
wacc	8.0%	we x ke + (1 - we) x kd x (1 - t)

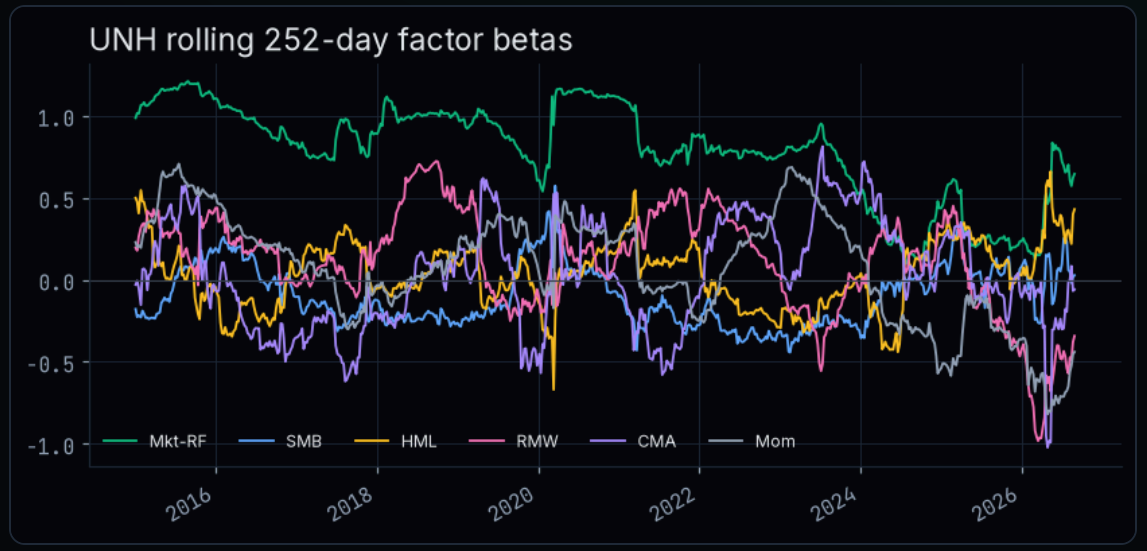


Base FCF $16.1B (FY2025 operating cash flow - capex (SEC 10-K)). Stage-1 growth 13.7% fades to 3.0% terminal growth over 10 years. Value $526.64. The price implies 6.0% growth.

EXPOSURES

Factor exposures

Factor	Beta	t-stat
Mkt-RF	+0.82	13.3
SMB	-0.12	-2.0
HML	+0.13	1.8
RMW	+0.26	4.3
CMA	+0.09	0.8
Mom	+0.08	1.7
Alpha (annual)	5.9%	0.8



FF5 plus momentum, daily, 2014-01-03 to 2026-08-31. R^2 0.23: most of UNH's risk is stock-specific. After the market, the profitability (RMW) tilt has the strongest t-stat. Alpha is not significant.

RISK

Risk profile

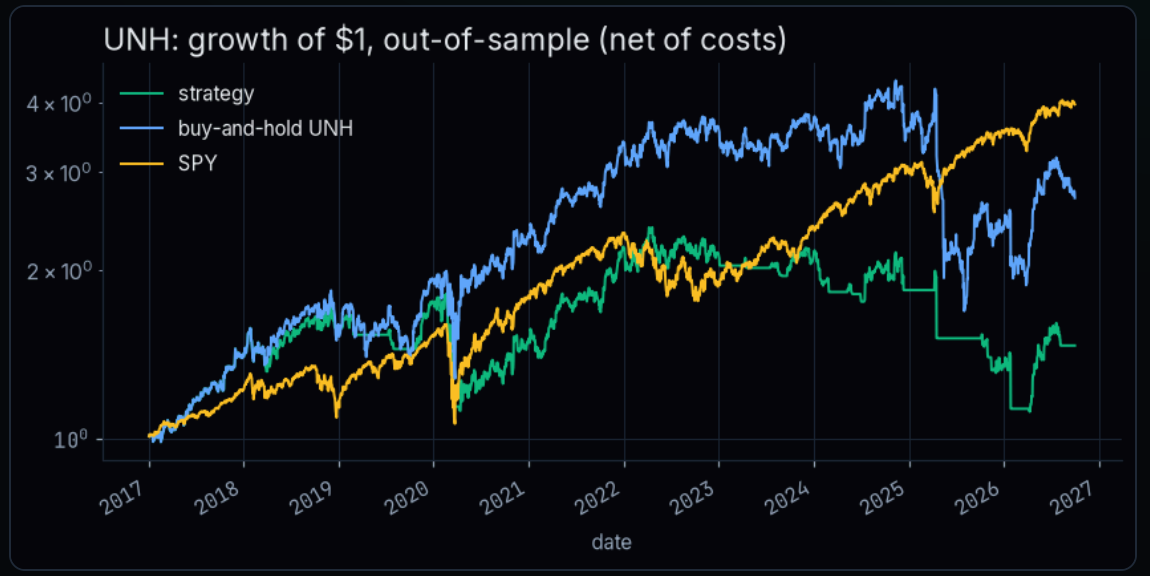
	Ann. vol	Beta vs SPY	Max drawdown	1-day 95% VaR	1-day 95% ES
Full sample: UNH	28.6%	0.78	-61.4%	2.3%	4.0%
Full sample: SPY	17.1%	1.00	-33.7%	1.6%	2.6%
Trailing 3y: UNH	36.8%	0.25	-61.4%	2.8%	5.8%
Trailing 3y: SPY	15.3%	1.00	-18.8%	1.4%	2.1%

UNH is more volatile than SPY and has suffered a -61.4% drawdown. Its low recent beta (0.25 trailing 3y) reflects stock-specific moves, not safety.

EVIDENCE

Backtested signal: it does not help

	CAGR	Sharpe (rf = 0)	Max DD
Trend signal	4.0%	0.29	-53.3%
Buy-and-hold UNH	10.8%	0.49	-61.4%
SPY	15.2%	0.87	-33.7%



Out-of-sample `2017-01-03` to `2026-09-30`, walk-forward window choice, next-day execution, `10` bp per switch. Both the signal and buy-and-hold UNH trail SPY on Sharpe. This is a valuation call, not a momentum trade.

SCENARIOS

Bull, base and bear

	Stage-1 growth	WACC	Value	vs price
Bear	7.6%	8.5%	$354.47	-3.4%
Base	13.7%	8.0%	$526.64	+43.5%
Bull	19.4%	7.5%	$764.68	+108.3%



The bear case is -3.4%, below today's price. Growth is set at the lower quartile, median and upper quartile of UNH's own rolling 5-year FCF growth, and WACC moves by ±0.5%. No probability weights are applied, because there is no data-based way to set them. All three cases still use the 5y beta that sets WACC at 8.0%. With the full-sample beta, the base case alone falls to -12.5% (slide 11).

CATALYSTS

What to monitor

- **Next 10-K FCF.** FY2025 FCF was **$16.1B**. The price implies **6.0%** a year from here. A further decline would break the thesis.
- **Operating margin.** It is **4.2%** against **8.7%** in FY2023. A recovery is the upside driver.
- **Beta re-estimate.** If the 5y beta drifts toward the full-sample **0.78**, WACC rises to **10.5%**.

No dated catalysts are listed: this project uses only the frozen data and does not forecast events.

RISKS

Key risks and what would change my mind

- **The bear case loses money.** The bear value of `$354.47` is `-3.4%` vs the price.
- **The discount rate rests on a low beta.** The 5y beta of `0.41` gives the `8.0%` WACC. The full-sample beta of `0.78` gives a `10.5%` WACC and a value of `$321.29` (`-12.5%`). *On that beta the long case disappears.*
- **FCF is falling.** It fell from `$25.7B` (FY2023) to `$16.1B` (FY2025), while stage-1 growth uses the `13.7%` long-run median.
- **The track record is weak.** Buy-and-hold UNH's Sharpe of `0.49` trails SPY's `0.87`, with a `-61.4%` max drawdown.

> **I would drop the long** if the next 10-K shows FCF still falling, or if a re-estimated beta pushes WACC to the point where the base value falls below the price.

METHOD

Methodology and reproducibility

- As-of `2026-09-30`. Data is frozen with SHA-256 hashes: Yahoo Finance prices and share counts, SEC EDGAR XBRL 10-K facts (filed on or before the as-of date), the Ken French library (factors end `2026-08-31`), and the FRED 10-year yield.
- Notebooks `02` to `06` produce every figure. `results/summary.json` holds every number. A test checks each number on these slides against that file.
- `make all` rebuilds everything offline and runs the tests.

APPENDIX

Appendix A: rejected candidates

	Price	DCF value	vs price	Bear	Bull	Price-implied growth
COST	$910.34	$410.96	-54.9%	$284.63	$496.94	38.5%

V (first pick, archived). First screen pick. The base DCF was well below the price, the bull case was also below the price, and the price implied more FCF growth than V's history. See `results/archive/V_summary.json`.

COST (third on the shortlist). Every scenario, including bull at $496.94, is below the price. The price implies 38.5% stage-1 growth.

V, UNH and COST use the same code and the same assumptions.

APPENDIX

Appendix B: assumptions

terminal growth	Assumption: roughly long-run nominal US GDP growth; kept below the 10y yield
fade years	Assumption: growth fades linearly from stage-1 to terminal over 10 years
scenario wacc shift	Assumption: +/-0.5pp WACC in bear/bull
trading cost bps	Assumption: 10 bp per one-way switch (commission plus spread for a mega-cap)
sma grid	Assumption: candidate trend-filter windows (trading days)
train years	Assumption: walk-forward training window
cash return	Assumption: cash earns 0 in the backtest (conservative for the strategy)
beta window years	Convention: 5y of daily returns for the CAPM beta